

LESSON 9-1 • SESSION 1 • PRACTICE • FORM B

Independent and Dependent Variables

Learning target – I can name the independent and dependent variables in a relationship. · 6.EE.C.9

REMEMBER

- The **independent variable** is the input: you choose it, or it comes first.
- The **dependent variable** is the output. It changes because of the input.
- Sentence test: “**dependent** depends on **independent**.” Independent goes on the x-axis.

WORKED EXAMPLE

A bike rental costs \$5 for each hour. Name the variables.

- Step 1** The quantities are hours rented and total cost.
Step 2 Cost changes when hours change.
Step 3 “Total cost depends on hours rented” makes sense.

Answer hours = independent; cost = dependent

WORDS independent variable (*variable independiente*) · dependent variable (*variable dependiente*) · depends on (*depende de*)

PRACTICE

show your work · label units

1. For each pair, write **I** under the independent variable and **D** under the dependent variable.

	Quantity	I or D	Quantity	I or D
a.	pages read		minutes reading	
b.	cups sold		money collected	
c.	calories burned		minutes jogged	
d.	players on a team		water bottles needed	

2. **Lemonade Stand.** Ava sells lemonade. The more cups she sells, the more money she collects. Hotter days lead to more cups sold.

- a. Cups sold and money collected. Independent: _____ Dependent: _____
 b. Temperature and cups sold. Independent: _____ Dependent: _____
 c. Which quantity is independent in one pair and dependent in the other? _____

3. **Plant Light.** Lucia gives each plant a different number of hours of light. Then she counts how many leaves grow.

Hint What does the student control? What is only measured?

- a. Independent variable: _____
 b. Dependent variable: _____

4. **Garden Hose.** The longer a hose runs, the more water fills a pool. A student says the time the hose runs is the independent variable. Do you agree?

Hint Use the sentence test: what depends on what?

Agree? _____

5. **Mowing Lawns.** Andre mows lawns. The more lawns he mows, the more gas he uses. Name the dependent variable.

Hint Which one changes because of the other?

Dependent _____

6. **Which Axis?** A graph shows the height of a candle and the hours it has burned.

Hint The independent variable goes across.

a. Which variable goes on the x-axis? _____

b. Which variable goes on the y-axis? _____

7. **Charging a Phone.** The longer Elena's phone is plugged in, the higher its battery percentage. Write a sentence with "depends on." Then name the dependent variable.

Hint Elena controls the time; the battery only responds.

Sentence _____

8. **Muffin Sale.** Every muffin costs the same price. A student says the number of muffins sold is the dependent variable because it is the smaller number. Explain whether you agree. Then name both variables.

Show your work.

Variables _____

FIND THE ERROR

explain the mistake, then fix it

9. **Jamal's Work.** Jada practices free throws. Jamal wrote: "The minutes Jada practices depend on her free-throw score." Explain his mistake and write a correct sentence.

TEST PRACTICE

choose the best answer

10. The dependent variable goes on the –

A x-axis

B y-axis

C origin

D title

11. In "gallons of fuel and miles driven," the dependent variable is –

A miles driven

B gallons of fuel

C both

D neither

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

cups sold D	plants Z	gas E	leaves E	hours N	Disagree Q	phone J	battery D	lawns X	Agree P
		#2c	#3b	#4	#5	#6a	#7		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____